LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: Carbon and its Compounds

Why carbon is special

- **Covalent bonds:** carbon (valency 4) shares electrons, because gaining or losing 4 electrons is very hard.
- **Catenation:** carbon bonds with other carbon atoms to form chains, branches and rings. **Tetravalency:** it can bond with 4 other atoms.
- Covalent compounds have low melting and boiling points and are poor conductors (no ions are formed).
- **Allotropes:** diamond (each C bonded to 4 others; hardest), graphite (layers; conducts electricity), fullerene C_{60}.

Hydrocarbons & naming

Series	General formula	Bond	Examples
Alkane (saturated)	C_nH_{2n+2}	single	methane CH_4, ethane C_2H_6
Alkene	C_nH_{2n}	double	ethene C_2H_4, propene C_3H_6
Alkyne	C_nH_{2n-2}	triple	ethyne C_2H_2, propyne C_3H_4

Homologous series: same functional group; consecutive members differ by $–CH_2–$ (14 u); similar chemical properties; physical properties change gradually. **Isomers:** same molecular formula, different structure (butane has 2).

Functional group	Formula	Name ending / prefix	Example
Halogen	–Cl, –Br	chloro-, bromo-	chloropropane
Alcohol	–OH	-ol	propanol
Aldehyde	–CHO	-al	propanal
Ketone	>C=O	-one	propanone
Carboxylic acid	–COOH	-oic acid	propanoic acid

Chemical properties

- **Combustion:** $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat and light. Saturated: clean blue flame. Unsaturated: sooty yellow flame.
- **Oxidation:** alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$ oxidises ethanol to ethanoic acid.
- **Addition (hydrogenation):** unsaturated + H_2, with Ni or Pd catalyst. Vegetable oil → vegetable ghee.
- **Substitution:** $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (in sunlight)

Ethanol & ethanoic acid

- Ethanol + sodium: $2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$
- Ethanol + hot concentrated H_2SO_4 at 443 K → ethene: $C_2H_5OH \rightarrow C_2H_4 + H_2O$
- Ethanoic acid: vinegar is a 5–8% solution. Pure acid freezes in winter (glacial acetic acid).
- Esterification: $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$ (acid catalyst; sweet smell)
- Saponification: $CH_3COOC_2H_5 + NaOH \rightarrow C_2H_5OH + CH_3COONa$
- With a carbonate: $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$

Soap: sodium or potassium salt of a long-chain carboxylic acid. Its ionic end is hydrophilic and its hydrocarbon tail is hydrophobic, so it forms **micelles** around oily dirt.
In hard water (Ca^{2+}, Mg^{2+}), soap forms insoluble **scum**. Detergents work in hard water too.

Quick check

1. How many covalent bonds are there in ethane, C_2H_6?

2. Name the functional group in propanone and in propanoic acid.

3. Why do unsaturated hydrocarbons burn with a sooty flame?